

such as those found in Facebook's RAG and Google's T5, are rapidly gaining traction in use cases requiring dynamic, context-aware generation.

Advantages of open source RAG agents are:

Improved accuracy: By retrieving contextually relevant information, RAG agents can provide more accurate and coherent responses, particularly in applications like chatbots, legal research, and technical support.

Real-time knowledge integration: Open source RAG agents can be seamlessly integrated with real-time data sources, making them ideal for applications in news aggregation, live customer service, and market analysis.

Applications of open source generative AI in real-world scenarios

Open source generative AI models are already having a profound impact across various industries. Some notable applications include:

Healthcare: The healthcare industry has witnessed a significant transformation with the application of open source generative AI. One of the key areas where these models are being utilised is clinical data analysis. Open source language models (LLMs) like GPT-Neo, GPT-3, and domain-specific variations of these models are used to sift through vast amounts of clinical data, medical literature, and patient records. By processing this data, these AI models can generate actionable insights, suggest diagnoses, and even predict patient outcomes, which aids healthcare professionals in making more informed decisions.

For example, an AI model can review a patient's medical history, understand symptoms described in the consultation, and recommend a series of diagnostic tests. Additionally, open source models enable the automation of tasks like medical

transcription. Speech-to-text models such as Mozilla's DeepSpeech or Kaldi help transcribe physicians' verbal notes, reducing the time spent on administrative tasks and improving accuracy. This is especially crucial in settings where real-time documentation is needed, such as during patient exams or surgeries. These models also improve accessibility by enabling real-time translation of medical information across different languages, facilitating communication between healthcare providers and patients from diverse linguistic backgrounds.

Moreover, AI-driven personalised treatment plans are becoming more achievable through generative models. These models, trained on patient data, can suggest personalised drug regimens based on factors such as medical history, genetic information, and lifestyle, ensuring that treatments are tailored to the unique needs of each patient.

Education: Open source generative AI models are helping create personalised learning experiences. Tools like GPT-3, fine-tuned for educational purposes, can engage with students in real-time, answer questions, and provide tailored learning paths based on a student's strengths and weaknesses. These models are being used in AI tutoring systems where they assist students with homework, explain complex concepts, and help reinforce learning materials. Such tutoring systems can be deployed as chatbots or virtual assistants, providing students with immediate feedback and support outside of traditional classroom hours.

Open source models also play a critical role in adaptive learning systems. By analysing student responses, AI models can modify the curriculum in real time to match the student's progress. This technology is not only valuable in traditional K-12 settings but also in higher education,

especially in online learning environments, where personalisation is key to student success. Moreover, AI-powered grading tools are streamlining the assessment process. These systems can assess essays and written content with high accuracy, freeing up time for educators to focus on more complex tasks such as providing feedback and mentoring.

Additionally, open source generative AI is making learning more accessible by providing real-time translation and transcription services, enabling students with hearing impairments or those who speak different languages to participate fully in educational environments. Tools like Google's T5 (Text-to-Text Transfer Transformer) model have been implemented for translating learning materials into multiple languages, fostering inclusivity in global classrooms.

Content creation and marketing: The content creation industry has been one of the earliest adopters of open source generative AI, and its impact is undeniable. Open source language models such as GPT-3 and GPT-Neo are now being used for automated content generation, including articles, blogs, social media posts, and even marketing copy. These models can write engaging and coherent content at scale, helping marketers and content creators maintain a consistent online presence without having to invest significant amounts of time in manual writing. For instance, AI models are generating product descriptions, email marketing campaigns, and even video scripts, tailored to the tone and style of the brand.

Generative AI is also transforming the visual content creation process. Models like DALL-E and CLIP (Contrastive Language-Image Pretraining) can generate images from textual